Beyond the hat model: the four hats without changing the plane, and the +3 question

Draft for discussion

20 October 2026

1 Beyond the hat model

Write $n_b(\Delta, 3)$ for the largest order of a bipartite graph of maximum degree Δ and diameter 3. For a prime power q, the construction of [1] gives $n_b(q+2, 3) \geq 2(q^2+q+5)$ and the four-hat construction gives $n_b(q+2, 3) \geq 2(q^2+q+6)$, two vertices more. Here we ask whether one more vertex is possible, that is, whether $n_b(q+2, 3) \geq 2(q^2+q+6)+1$ ("+3" over [1]). We cannot settle this. We record where the question is open, what any such graph must look like when it is built on a projective plane of order q and q is large, and what a computer search covers.

1.1 Where the question is open

The bipartite Moore bound for diameter 3 is $2(\Delta^2 - \Delta + 1)$; for $\Delta = q+2$ it is $2(q^2+3q+3)$, which exceeds $2(q^2+q+6)$ by $4q-6$. It is attained if and only if a projective plane of order $\Delta - 1 = q+1$ exists. When both q and $q+1$ are prime powers (one of them is then a power of 2: $q \in \{2, 3, 4, 7, 8, 16, 31, \dots\}$), $\mathrm{PG}(2, q+1)$ attains it and our question is void. Bipartite graphs of defect 1 or 3 do not exist (for defect less than Δ such graphs are regular, hence of even order; see [**?**, Prop. 3.1]), and the defect-2 case at diameter 3 is restricted by [2]. None of these upper bounds comes near: for $q \geq 9$ the value $2(q^2+q+6)+1$ has defect $4q-7 \geq 29$.

In the range of the Combinatorics Wiki table ($\Delta \leq 16$) the relevant cells are:

Δ	q	[1]	four hats	upper bound (table)	$2(q^2+q+6)+1$
7	5	70	72	80	$73 < 80$
11	9	190	192	220	193; open
13	11	274	276	314	277; open
15	13	374	376	422	377; open

At $\Delta = 7$ the value $n_b(7, 3) = 80$ is known exactly [3]: the table has a graph of order 80, and [3, Thm. 4.1] rules out bipartite $(7, 3, -4)$-graphs, the last case left. For $\Delta = 16$ neither $\Delta - 1 = 15$ nor $\Delta - 2 = 14$ is a prime power, so the question does not arise. Beyond the table, $2(q^2+q+6)+1$ is the first open value for every prime power $q \geq 9$ with $q+1$ not a prime power ($q = 9, 11, 13, 17, 19, 23, 25, 27, 29, 32, 37, \dots$), as far as we know.

1.2 Extensions of a projective plane

Every graph in §**??** contains the incidence graph of $\mathrm{PG}(2, q)$ with a few incidences changed. We now ask what any graph of this kind must look like.

Let Π be a projective plane of order q with point set $\mathcal{P}$ and line set $\mathcal{L}$, and $N = q^2 + q + 1$. An *extension* of Π is a bipartite graph G with parts $\mathcal{P} \cup P'$ (points) and $\mathcal{L} \cup L'$ (lines), of maximum degree at most $q + 2$ and diameter at most 3. The vertices in P' and L' are *new*, the others *old*. A *change* is a pair (old point, old line) on which G and Π disagree: a *deleted* flag or an *added* non-incident pair. Write $p = |P'|$, $l = |L'|$ and e for the number of changes. The order of G is $2N + p + l$; so [1] has $p = l = 4$, the four-hat graph has $p = l = 5$, and "+3" asks for $p + l = 11$. In a bipartite graph without isolated vertices, diameter at most 3 means: any two points have a common neighbour and any two lines have a common neighbour (a point x and a line $L \not\ni x$ are then at distance 3 through any point of L).

For a new point h, let $O(h)$ be the set of old lines through h, $a_h = |O(h)|$, and d_h the number of new lines through h. Dually $O(K)$, c_K for a new line K. For an old point x let $r_x(h)$ be the number of lines of $O(h)$ that pass through x *in* Π, and $\sigma_h = \sum_x \max(r_x(h) - 1, 0)$.

Lemma 1.1. (a) *An old point x lies on at most $1 + \delta(x) - \alpha(x)$ new lines, where $\delta(x)$ and $\alpha(x)$ count the deleted and added pairs at x. Dually for old lines.*

(b) *Two new points with no common new line lie on a common old line carrying a deleted flag. In particular, if $e = 0$, any two new points lie on a common new line, and any two new lines meet in a new point.*

Proof. (a) x has $q + 1 - \delta(x) + \alpha(x)$ old neighbours and degree at most $q + 2$. (b) By (a), an old line through two new points has a deleted flag. □

Lemma 1.2. *Let h be a new point, $a = a_h$, $d = d_h$, and let t be the number of added pairs (x, L) with $L \in O(h)$. Then $\sigma_h \le (a + d)(q + 1) - N + t \le 2q + 1 + t$, and $a + d \ge q + 1 - \varepsilon$, where $\varepsilon = 0$ if $t = 0$ and $\varepsilon = 1$ if $1 \le t \le q + 1$.*

Proof. In Π the a lines of $O(h)$ cover $a(q + 1) - \sigma_h$ points; in G they cover at most t more. Every other old point is at distance 2 from h through a new line, and a new line through h has at most $q + 1$ old points. Hence $N - a(q + 1) + \sigma_h - t \le d(q + 1)$. Use $a + d \le q + 2$ for the first claim and $\sigma_h \ge 0$ for the second. □

Lemma 1.3 (near-copies)**.** *Let h, a, t, ε be as in Lemma 1.2 and suppose $q \ge 3l + 6 + t + 3\varepsilon$ (so $t \le q - 6$ and ε is defined). Then there is an old point X such that all but at most one line of $O(h)$ pass through X (in Π). At least $q - l - \varepsilon - 1$ lines of $O(h)$ pass through X.*

Proof. Let X maximise $r = r_X(h)$, and let $b = a - r$ be the number of lines of $O(h)$ not through X; put $s = 2q + 1 + t$, so $\sigma_h \le s$ by Lemma 1.2. Two facts: (i) $(b + 1)r \le s + 1$. Indeed X contributes $r - 1$ to σ_h; every point $x \ne X$ on one of the r lines through X lies on exactly one of them and contributes the number of non-X lines through it; and each of the b non-X lines meets each of the r lines in a point other than X. So $\sigma_h \ge r - 1 + br$. (ii) $a(a - 1) \le rs$. Any two lines of $O(h)$ meet in one point of Π, so $a(a - 1) = \sum_x r_x(r_x - 1) \le r \sum_x \max(r_x - 1, 0) = r\sigma_h$.

By Lemma 1.2, $a \ge q + 1 - l - \varepsilon$, and the hypothesis gives $a \ge (2q + 9 + t)/3 = (s + 8)/3$. Suppose $b \ge 2$. Then $r \le (s + 1)/3$ by (i), $r \ge r_0 := a(a - 1)/s$ by (ii), and $a = r + b \le g(r) := r + (s + 1)/r - 1$ by (i). As g is convex, $a \le \max\{g(r_0), g((s + 1)/3)\}$. First, $g((s + 1)/3) = (s + 7)/3 < a$. Second, $a/3 < r_0 \le a(q + 1)/(2q + 1)$ (use $a \ge (s + 5)/3 + 1$ and $a \le q + 2$), and $x \mapsto x + (s + 1)/x$ is convex, so $g(r_0) + 1$ is at most the larger of $a/3 + 3(s + 1)/a < a/3 + 9$ and $a(q + 1)/(2q + 1) + (s + 1)(2q + 1)/(a(q + 1)) < a(q + 1)/(2q + 1) + 6$. Both are at most $a + 1$ once $a \ge 13$, which holds for $q \ge 15$. So $g(r_0) < a$, a contradiction. For $q \le 14$ the hypothesis forces $l \le 2$ and $t \le 5$, and an exhaustive check of the integer system (i), (ii), $a = r + b$, $b \ge 2$, $q + 1 - l - \varepsilon \le a \le q + 2$ finds no solution

(`thresholds.py`, which checks every q from the threshold up to 400 for $0 \le l \le 12$, $t \le 12$; the bound $3l + 6 + t + 3\varepsilon$ is exactly the threshold of this system in every case checked). Hence $b \le 1$ and $r = a - b \ge q - l - \varepsilon - 1$. □

We call X the point *copied* by h and write $X = X_h$; dually a new line K copies an old line L_K. In the four-hat graph, Q copies A, H_s copies X_s, M copies ℓ and K_s copies AX_s.

Theorem 1.4. *Let G be an extension of Π with p new points, l new lines and e changes, and suppose $q \ge 3\max(p, l) + 9 + e$. Then:*

(a) *every new point h copies an old point X_h: all but at most one of its old lines pass through X_h, and it has at least $q - l - 2$ old lines through X_h; dually every new line K copies an old line L_K;*

(b) *distinct new points copy distinct points, and distinct new lines copy distinct lines;*

(c) *for every line M of Π through X_h that is not an old line through h, some new line K through h copies M. If m_h is the number of such lines M, then $m_h \le d_h \le m_h + 1$, and $d_h = m_h$ if h lies on an old line not through X_h. Dually for new lines.*

Proof. (a) is Lemma 1.3 with $t \le e$ and $\varepsilon \le 1$, and its dual. (b) If $X_h = X_{h'}$ for new points $h \ne h'$, then at least $2(q - l - 2) - (q + 1)$ lines through X_h are old lines through both h and h'; each of them carries a deleted flag by Lemma 1.1(a), so $q - 2l - 5 \le e$, against the hypothesis. Dually for lines. (c) The r old lines of h through X_h meet M only in X_h, the possible extra old line of h meets it in one point, and added pairs put at most e further points of M on old lines of h. So at least $q - 1 - e$ points of M are at distance 2 from h only through new lines through h. A new line K through h with $L_K \ne M$ has at most two old points on M (one on L_K, and possibly its one old point off L_K). If no K through h copied M, then $q - 1 - e \le 2d_h \le 2l$, which is false. Different lines M have different copies, so $d_h \ge m_h$. Finally $a_h = (q + 1 - m_h) + b$ with $b \in \{0, 1\}$ the number of old lines of h not through X_h, and $a_h + d_h \le q + 2$. □

So, for large q, every graph of order $2N + k$ built on a projective plane by e changes is a *copy configuration*: a set $\mathcal{X}$ of copied points and a set $\mathcal{L}'$ of copied lines, such that every line through a copied point is either an old line of its copy or is itself copied (and that copy passes through the point's copy), and dually. Each new vertex has at most one "free" new neighbour that is not explained by this rule. This is the general form of the hat lemma ([1]; §??): in the four-hat graph $\mathcal{X} = \{A, X_0, X_1, X_a, X_c\}$ and $\mathcal{L}' = \{\ell, AX_0, AX_1, AX_a, AX_c\}$; the hats H_0, H_1, H_c have $m = 1$ (their missing line ℓ is copied by M), H_a lies on ℓ and has $m = 0$, and Q has $m = 3$ (missing AX_0, AX_1, AX_c, copied by K_0, K_1, K_c; its free line is K_a). The hat model of the SAT search in §?? fixes this particular configuration. Theorem 1.4 says that for large q only copy configurations occur; the next proposition shows that without changes the configuration must have the shape of the hat configuration.

Proposition 1.5. *Let G be an extension of Π with no changes ($e = 0$), $q \ge 3\max(p, l) + 9$ and $p + l \le 13$. Then the line joining any two copied points is copied, and the point where any two copied lines meet is copied. Consequently all copied points but at most one lie on one line, and all copied lines but at most one pass through one point.*

Proof. Let $h \ne h'$ be new points and $J = X_h X_{h'}$ (the points are distinct by Theorem 1.4(b)). An old line has at most one new point when $e = 0$ (Lemma 1.1(a)), so J is not an old line of both h and h'; if it is not an old line of h, Theorem 1.4(c) says J is copied. The second claim is dual.

So $(\mathcal{X}, \mathcal{L}')$ is closed under joins and meets. If $\mathcal{X}$ contained four points with no three collinear, the closure would be a subplane, with at least 7 points and 7 lines, against $p + l \leq 13$. A point set without such a quadrangle has all but at most one of its points on one line: otherwise take a line L with the most points of the set, $m \geq 2$ of them, and two points u, v of the set off L; if $m \geq 3$, two points of L off the line uv together with u, v form a quadrangle, and if $m = 2$ any four points do. Dually for $\mathcal{L}'$. □

With changes the same argument shows that at most e lines joining two copied points can fail to be copied (such a line is an old line of both copies, so it carries a deleted flag, and distinct lines carry distinct flags); dually for meets. So the configuration is closed up to e exceptions. In the hat configuration the line is ℓ, the point is A, and $A \notin \ell$.

Theorem 1.6. *Let G be an extension of Π with no changes, p new points and l new lines, $p+l \leq 13$ and $q \geq 3\max(p, l) + 9$. Then $p \leq 6$ and $l \leq 6$, and $p = 6$ or $l = 6$ forces $p = l = 6$. In particular $p+l \notin \{11, 13\}$: for $q \geq 42$ (respectively $q \geq 48$) no graph of maximum degree $q+2$ and diameter 3 consists of the incidence graph of a projective plane of order q as an induced subgraph plus exactly 11 (or 13) further vertices.*

Proof. By Proposition 1.5 there is a line L^* containing all copied points but at most one; let $S = \mathcal{X} \cap L^*$ and $s = |S| \geq p - 1$. Suppose $s \geq 6$. Then L^* is copied (it joins two copied points), say by K^*; write h_Y for the new point copying $Y \in S$.

(i) L^* has at most one new point (Lemma 1.1), so it is an old line of h_Y for at most one $Y \in S$. For every other Y, L^* is missing at h_Y, so K^* passes through h_Y (Theorem 1.4(c)).

(ii) The new points on K^* are the copies of the points of L^* that are not old points of K^*, and at most one more (the dual of Theorem 1.4(c)). Hence at most two points $Y \in S$ are old points of K^*: the one with $L^* \in O(h_Y)$, and one whose copy is the extra new point of K^*.

(iii) A new line $K \neq K^*$ contains at most two copies h_Y, $Y \in S$: its new points are copies of points of L_K that are not old points of K, plus at most one more; $L_K \neq L^*$ meets L^* in one point, so at most one of the former copies a point of S.

(iv) Let $Y \neq Y'$ in S with L^* missing at $h_{Y'}$. The old lines of $h_{Y'}$ are lines through Y' other than $L^* = YY'$, and at most one further line $E_{Y'}$. So unless $Y \in E_{Y'}$, the points Y and $h_{Y'}$ are joined only by a new line, which must be the unique new line $\nu(Y)$ through Y.

Let e_Y be the number of $Y' \in S$ with $Y \in E_{Y'}$; as each $E_{Y'}$ meets L^* once, $\sum_Y e_Y \leq s$. By (i) and (iv), at least $s - 2 - e_Y$ copies $h_{Y'}$ must lie on the new line $\nu(Y)$ (if this number is positive, $\nu(Y)$ exists). For the at least $s-2$ points Y that are not old points of K^* (by (ii)), $\nu(Y) \neq K^*$, so by (iii) $s-2-e_Y \leq 2$, that is $e_Y \geq s-4$ (also when $\nu(Y)$ does not exist). Then $(s-2)(s-4) \leq \sum_Y e_Y \leq s$, which fails for $s \geq 6$. So $s \leq 5$ and $p \leq 6$. If $p = 6$, then $s = 5$ and there is a copied point $u \notin L^*$; the five lines uY, $Y \in S$, are copied (Proposition 1.5) and so is L^*, so $l \geq 6$. The statements for l are dual. If $p + l \in \{11, 13\}$ then $\max(p, l) \geq 6$ and $\min(p, l) \leq 6$ with $(p, l) \neq (6, 6)$, a contradiction. □

The hypothesis $q \geq 3\max(p, l) + 9$ is what Theorem 1.4 needs; the counting in the proof itself does not depend on q. The proof also survives a single added pair and no deleted flag (with $q \geq 3\max(p, l) + 10$): without deletions the closure is exact and every old vertex still has at most one new neighbour, and the added pair raises $\sum_Y e_Y$ by at most one, while $(s-2)(s-4) \leq s+1$ still fails at $s = 6$. The slack at $s = 6$ is only two, so two added pairs, or one deleted flag (which

lets one old point carry two new lines and one old line two new points), already break the count. With changes the argument breaks in two places: a changed incidence can let two new points share an old line, and can let a copied point see a hat through an added old incidence. So a $+3$ on a projective plane of large order must change the plane, and Theorem 1.6 says nothing about how many changes would suffice. Whether $p = l = 6$ (that is, $2(q^2+q+7)$, five hats) can occur without changes is also left open by the counting; the hat-model SAT search of §?? found no such graph.

1.3 The four hats without changing the plane

The four-hat construction of §?? moves three incidences of the plane (the rotation). The computer search below found that this is not necessary: the rotation can be carried by the copies.

Proposition 1.7. *Let Π be a projective plane of order $q \geq 3$, ℓ a line, A a point not on ℓ, and X_a, X_1, X_2, X_3 distinct points of ℓ. Let s be the 3-cycle $1 \mapsto 2 \mapsto 3 \mapsto 1$ and write AX for the line through A and X. Add new points Q, H_a, H_1, H_2, H_3 and new lines M, K_a, K_1, K_2, K_3 to the incidence graph of Π, with these incidences ($i \in \{1,2,3\}$):*

vertex	*old neighbours*	*new neighbours*
Q	*every line through A except AX_1, AX_2, AX_3*	K_a, K_1, K_2, K_3
H_a	*every line through X_a except AX_a*	M, K_a
H_i	*every line through X_i except ℓ and AX_i; also $AX_{s(i)}$*	M, K_i
M	*every point of ℓ except X_1, X_2, X_3*	H_a, H_1, H_2, H_3
K_a	*every point of AX_a except X_a*	Q, H_a
K_i	*every point of AX_i except A and X_i; also $X_{s^{-1}(i)}$*	Q, H_i

The resulting graph G is bipartite, has maximum degree $q+2$, diameter 3 and order $2(q^2+q+6)$, and every incidence of Π is kept: the incidence graph of Π is an induced subgraph of G.

Proof. Degrees. New vertices: Q has $(q+1-3)+4$, H_a has $q+2$, H_i has $(q-1)+1+2$, M has $(q-2)+4$, K_a has $q+2$ and K_i has $(q-1)+1+2$ neighbours. Every old vertex gets at most one new neighbour: a line through A meets ℓ in one point, so the old lines with a new point are AX_a and the lines AP, $P \in \ell \setminus \{X_a, X_1, X_2, X_3\}$ (point Q), the lines through X_a other than AX_a, among them ℓ (point H_a), the lines through X_i other than ℓ and AX_i (point H_i), and AX_i (point $H_{s^{-1}(i)}$), and these classes are disjoint. Dually the old points with a new line are the points of $\ell \setminus \{X_1, X_2, X_3\}$ (line M), of $AX_a \setminus \{X_a\}$ (line K_a), of $AX_i \setminus \{A, X_i\}$ (line K_i), and X_i (line $K_{s(i)}$).

Two points. Old pairs are joined in Π. Q and A lie on AX_a. Q and an old point $x \neq A$: if $Ax \neq AX_i$ for all i then Ax is a line of Q; if $x \in AX_i \setminus \{A, X_i\}$, use K_i; if $x = X_i$, use $K_{s(i)}$. H_a and x: if $x \notin AX_a$ or $x = X_a$ use $X_a x$; otherwise use K_a. H_i and x: if $X_i x \notin \{\ell, AX_i\}$ use it; A and $X_{s(i)}$ lie on $AX_{s(i)}$; the points of ℓ other than X_1, X_2, X_3 lie on M; $X_{s^{-1}(i)}$ and the points of $AX_i \setminus \{A, X_i\}$ lie on K_i; and X_i is on every line of H_i through it. New pairs: Q, H_a on K_a; Q, H_i on K_i; any two hats on M.

Two lines. M and an old line L: if L meets ℓ in a point of M, done; if $L \ni X_i$ and $L \neq \ell$, then L contains H_i (when $L \neq AX_i$) or $H_{s^{-1}(i)}$ (when $L = AX_i$), and both lie on M. K_a and L: L meets AX_a; if only in X_a, then L is a line through X_a other than AX_a and contains H_a. K_i and AX_i share the $q-1 \geq 2$ points of $AX_i \setminus \{A, X_i\}$. K_i and $L \neq AX_i$: let $y = L \cap AX_i$. If $y \notin \{A, X_i\}$, done. If $y = X_i$ and $L \neq \ell$, L contains H_i; if $L = \ell$, it contains $X_{s^{-1}(i)} \in K_i$. If $y = A$, then L contains Q unless $L = AX_j$ with $j \neq i$; then either $j = s(i)$ and L contains H_i, or $j = s^{-1}(i)$ and L contains $X_j \in K_i$. New pairs: M meets K_a in H_a and K_i in H_i; the K's meet in Q. □

The program `construction0.py` builds G from the table alone and checks it on random frames for $q = 3, 4, 5, 7, 8, 9, 11, 13, 16$ and on the nearfield plane of order 9; the graphs for $q = 4, 5, 9, 11, 13$ pass `verify_record.py`, and removing either rotation incidence ($H_i \sim AX_{s(i)}$ or $X_{s^{-1}(i)} \sim K_i$) breaks the diameter. In the language of Theorem 1.4: H_1, H_2, H_3 each miss two lines through their point (ℓ and AX_i, copied by M and K_i) and carry one old line not through it ($AX_{s(i)}$); K_1, K_2, K_3 are dual. An AL-size graph without changes ($p = l = 4$: Q, M and three hats whose rotation is a 3-cycle; Q and M each carry one extra old incidence) was found by the same search at $q = 7$ (`graphs/ext_q7_4_4_E0_A0_n122.json`).

1.4 Computer evidence beyond the hat model

The program `sat_ext.py` encodes extensions of $\mathrm{PG}(2, q)$ directly: p new points and l new lines with all their incidences free; at most E flags of the plane may be deleted and at most A non-incident old pairs added; degree at most $q + 2$; any two points and any two lines have a common neighbour. Nothing about hats, frames or gadgets is fixed. Symmetry is broken soundly: new points are sorted by their old lines and new lines by their old points, and the action of $\mathrm{P\Gamma L}(3, q)$ is used in one of two ways.

- *Full model.* Either no new point lies on two old lines (case A), or, for $r = 2, \ldots, q + 1$, r is the largest number of old lines through one old point that any new point lies on, attained by new point 0 at old point 0 on $\min(r, 3)$ fixed lines of its pencil (case r). These $q + 1$ cases cover everything.

- *Copy model.* The conclusions of Theorem 1.4 are imposed (a theorem for $q \geq 3\max(p, l) + 9 + e$, an assumption below that); new point 0 copies old point 0, and new line 0 copies a fixed line through it (case "on") or not through it (case "off"). Optionally ("changes in the configuration") changes are allowed only between copied points and copied lines, as in both known constructions. Deleted flags are taken maximal (a deleted flag with spare degree at both ends could be put back), which is sound.

Every satisfying assignment is decoded and checked independently; every SAT graph quoted below passes `lib.is_valid`. Runs used CaDiCaL 1.5.3 or 1.9.5 through PySAT, at most four in parallel, each with a wall-clock limit.

In the table, E and A bound the deleted flags and the added pairs; "all cases" means case A and every case r.

model	q	(p, l)	E, A	result
full	4	$(5,4)$, $(6,3)$	$0, 0$	UNSAT, all cases (so no +1 over [1] without changes at $q = 4$ for these splits)
full	4	$(5,5)$	$0, 0$	SAT; same copy configuration and 3-cycle of extra incidences as Proposition 1.7
full	4	$(6,5)$	$0, 0$	UNSAT, all cases (longest 166 s)
full	5	$(5,4)$	$0, 0$	UNSAT in case A and case $r = 2$ (389 s); cases $r \geq 3$ not finished
full	4	$(6,5)$	$8, 8$	unknown: every case hit the 600 s limit
copy	$5, 7, 9$	$(5,5)$	$0, 0$	SAT in both cases (controls)
copy	7	$(6,5)$	$0, 0$	UNSAT in case "off" (243 s); case "on" unknown (1500 s)
copy	9	$(6,5)$	$0, 0$	unknown in both cases (900 s)
copy	$7, 9$	$(6,5)$	$1, 1$	unknown: 900 s limit
five hats	7	$(6,5)$	see right	UNSAT for $E = A = 0, 1, 2, 3$ (longest 1048 s); $E = A = 4$: unknown (1500 s); $E = A = 8$: unknown (900 s)
five hats	9	$(6,5)$	see right	UNSAT for $E = A = 1, 2$ (longest 515 s); $E = A = 3$: unknown (1500 s); $E = A = 4, 8$: unknown (900 s)
five hats	11	$(6,5)$	$1, 1$	UNSAT (132 s); $E = A = 2$: unknown (900 s)
five hats	13	$(6,5)$	$1, 1$	UNSAT (493 s)
five hats	$5, 7, 9$	$(5,5)$	$4, 3$	SAT (controls, changes only between copied vertices; 4 deletions used)
five hats	7	$(5,5)$	$4, 3$	SAT (control, changes anywhere; 153 s)

The "five hats" rows use the copy model with the configuration fixed to the natural $(6, 5)$ shape: a closed configuration as in Proposition 1.5 with one join left uncopied (which needs a deleted flag on it): Q copies a point u, M copies a line $L^* \not\ni u$, five new points copy five points $Y_1, \ldots, Y_5$ of L^*, and four new lines copy $uY_1, \ldots, uY_4$. Three of the Y_i are fixed (the stabiliser of (u, L^*) is 3-transitive on L^*); everything else is free, and changes may be anywhere in the plane. This is the configuration of the hat-model case $(5, 4, 1, 1)$ of §**??**, but with all incidences of the new vertices free. (With changes restricted to pairs of copied vertices this shape is impossible for a trivial reason: the uncopied join uY_5 would carry two new points and needs a deleted flag. Those runs are not evidence and are not listed.)

Local search. The annealer of our 376-vertex program, which inserts a vertex on each side and then swaps any edges (incidences of the plane included), was started from the graph of Proposition 1.7 with a filler matching at $q = 9$: two runs of 300 s found no graph of order 194 (`grow0.py`). Earlier runs from the four-hat graph at $q = 7, 9$ also found none.

1.5 Summary

- The +2 does not need to change the plane (Proposition 1.7).

- For large q ($q \geq 3\max(p, l) + 9 + e$), every extension of a projective plane by p new points, l new lines and e changes is a copy configuration (Theorem 1.4).
- Without changes, +3 is impossible for $q \geq 42$ (Theorem 1.6); at $q = 4$ the computer shows it for the split $(6, 5)$, and at $q = 7$ in the copy model, case "off".
- With changes the question is open. The natural +3 shape (five hats, one hat line short) is UNSAT with up to two deletions and two additions anywhere at $q = 7, 9$ (three of each at $q = 7$) and with one of each at $q = 11, 13$, in the copy model; larger budgets did not finish. Other shapes (an uncopied L^*, which needs at least four deletions on it; six or more copied points on a line; configurations that are not closed because of changes) and the full model with changes were not settled.
- Nothing here gives a new table value: at the open cells $\Delta = 11, 13, 15$ the best known orders stay $192, 276, 376$.

References

[1] G. Araujo-Pardo and N. López, On new record graphs close to bipartite Moore graphs, Graphs and Combinatorics 38 (2022), no. 4, Art. 110, doi:10.1007/s00373-022-02500-3 (arXiv:2005.02427).

[2] C. Delorme, L. K. Jørgensen, M. Miller, G. Pineda-Villavicencio, On bipartite graphs of diameter 3 and defect 2, J. Graph Theory 61 (2009) 271–288.

[3] R. Feria-Purón, M. Miller, G. Pineda-Villavicencio, On large bipartite graphs of diameter 3, Discrete Math. 313 (2013) 381–390, arXiv:1203.3588.